

在[16]中...

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, GridSearchCV, RandomizedSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.datasets import fetch_california_housing, load_breast_cancer

# 评价指标
from sklearn.metrics import (
    mean_absolute_error, mean_squared_error, r2_score, accuracy_score, precision_score,
    recall_score, f1_score, roc_auc_score, roc_curve, confusion_matrix, classification_report
) # precision 精确率: 判断为正的样本, 里面是正的概率; recall 召回率: 真实的正样本里面

# 中文显示设置
plt.rcParams['font.sans-serif'] = ['SimHei', 'Arial Unicode MS']
plt.rcParams['axes.unicode_minus'] = False

# 固定随机种子, 保证结果可复现
RANDOM_STATE = 42
```

一. 回归任务

- 回归任务公共操作

在[17]中...

```
# 加载加州房价数据集
reg_data = fetch_california_housing()
X_reg = pd.DataFrame(reg_data.data, columns=reg_data.feature_names) # 特征
y_reg = reg_data.target # 标签

# 数据集划分
# 训练集80%, 测试集20%; 固定随机种子, 保证可复现
X_reg_train, X_reg_test, y_reg_train, y_reg_test = train_test_split(
    X_reg, y_reg, test_size=0.2, random_state=RANDOM_STATE
)

# 标准化
# 树模型(RF/XGB/LGBM/AdaBoost)不需要标准化, 但线性回归需要
# 在训练集上fit, 在测试集上transform, 避免数据泄露
scaler = StandardScaler()
X_reg_train_scaled = scaler.fit_transform(X_reg_train)
X_reg_test_scaled = scaler.transform(X_reg_test)
```

- 了解基础数据

在[18]中...

```
X_reg.head(3)
```

出局[18]...

	MedInc	HouseAge	AveRooms (房室)	AveBedrm	人口	AveCup	纬度	经度
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24

在[19]中...

```
y_reg[:3] #房价
```

出局[19]...

```
array([4.526, 3.585, 3.521])
```

1.线性回归

在[20]中...

```
from sklearn.linear_model import LinearRegression,Ridge #调参通过正则化实现

# 1. 训练基础模型
lr=LinearRegression() #采用OLS最小二乘法,不需要设置随机数种子
lr.fit(X_reg_train_scaled,y_reg_train) # 线性回归必须要用标准化后的特征
y_pred_lr=lr.predict(X_reg_test_scaled)

# 2. 调参方法
# 线性回归本身无超参数可调
# 实际工程中通过Ridge(L2正则化)/Lasso(L1正则化)来'调参'—调整正则强度alpha
ridge_params={'alpha':[0.001,0.01,0.1,1,10,100]} # 候选正则强度
ridge_grid=GridSearchCV(
    Ridge(random_state=RANDOM_STATE),
    ridge_params,
    cv=5, # 五折交叉验证
    scoring='neg_mean_squared_error', # 负均方误差,因为GridSearchCV默认是最大化
    n_jobs=-1
)
ridge_grid.fit(X_reg_train_scaled,y_reg_train)
y_pred_ridge=ridge_grid.predict(X_reg_test_scaled)
print(f'Ridge最优参数alpha:{ridge_grid.best_params_}')
best_lr=ridge_grid.best_estimator_ # 获取最优模型
y_pred_lr=best_lr.predict(X_reg_test_scaled) # 使用最优模型预测

# 3. 评价指标
mae_lr=mean_absolute_error(y_pred_lr,y_reg_test)
mse_lr=mean_squared_error(y_pred_lr,y_reg_test)
r2_lr=r2_score(y_reg_test,y_pred_lr)
print(f'线性回归:MAE={mae_lr:.4f},MSE={mse_lr:.4f},R2={r2_lr:.4f}')

# 可视化
fig=plt.figure(figsize=(14,5))
ax1=plt.subplot(121) # 第一幅图:预测值VS真实值散点图
ax2=plt.subplot(122) # 第二幅图:残差分布图(应近似随机分布)

# 第一幅图:预测值VS真实值散点图
sns.scatterplot(x=y_reg_test,y=y_pred_lr,ax=ax1,palette='viridis',alpha=0.3)
ax1.plot([0,5],[0,5],ls='--',color='r',lw=2,label='理想预测线') # 对角参考线
ax1.set_xlabel('真实房价')
ax1.set_ylabel('预测房价')
ax1.set_title('线性回归:预测值VS真实值')
ax1.legend()

# 第二幅图:残差分布图
```

```

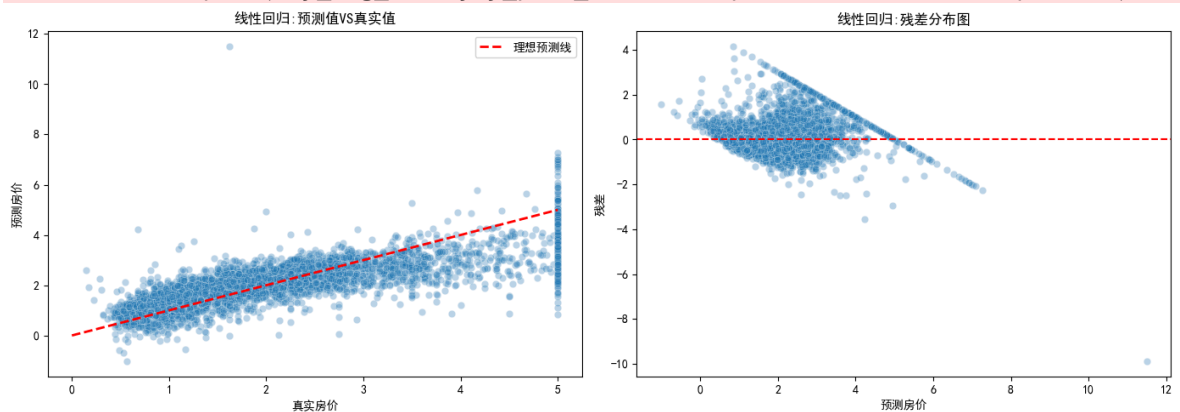
resid=y_reg_test-y_pred_lr
sns.scatterplot(x=y_pred_lr,y=resid,alpha=0.3,ax=ax2)
ax2.axhline(y=0,color='r',ls='--') # 残差为0的参考线
ax2.set_xlabel('预测房价')
ax2.set_ylabel('残差')
ax2.set_title('线性回归:残差分布图')

plt.tight_layout()

```

Ridge最优参数alpha: {'alpha': 0.1}
 线性回归: MAE=0.5332, MSE=0.5559, R2=0.5758

C:\Users\11146\AppData\Local\Temp\ipykernel_36404\270937261.py:37: UserWarning: Ignoring `palette` because no `hue` variable has been assigned.
 sns.scatterplot(x=y_reg_test,y=y_pred_lr,ax=ax1,palette='viridis',alpha=0.3)



- 结论: 标签房价和部分特征存在非线性关系难以拟合, 同时存在异常值/极端值干扰模型

2. 随机森林

在[21]中...

```

from sklearn.ensemble import RandomForestRegressor

# 1. 训练基础模型(树模型不需要标准化, 直接用原始特征)
rf_reg=RandomForestRegressor(n_estimators=100,n_jobs=-1,random_state=RANDOM_STATE)
rf_reg.fit(X_reg_train,y_reg_train)

# 2. 调参方法(RandomizedSearchCV随机搜索)
# 关键超参数说明:@min_samples_split: 节点分裂所需最小样本数 @min_samples_leaf: 叶子节点最小样本数
rf_params={
    'n_estimators':[100,200,300,500], # 树的数量
    'max_depth':[None,10,15,20], # 树的最大深度
    'min_samples_split':[2,5,10], # 节点分裂所需最小样本数
    'min_samples_leaf':[1,2,4], # 叶子节点最小样本数(层优先)
    'max_features':['sqrt','log2'] # 每棵树随机选择的特征数
}

rf_search=RandomizedSearchCV(
    RandomForestRegressor(n_jobs=-1,random_state=RANDOM_STATE),
    rf_params,
    n_iter=20, # 随机搜索20次
    cv=5, # 五折交叉验证
    scoring='neg_mean_squared_error', # 负均方误差
    n_jobs=-1,
    random_state=RANDOM_STATE,
    verbose=2 # 打印两个关键信息
)

rf_search.fit(X_reg_train,y_reg_train)
print(f'RF最优参数:{rf_search.best_params_}')

```

```

best_rf=rf_search.best_estimator_
y_pred_rf=best_rf.predict(X_reg_test)

# 3. 评价指标
print(f'随机森林:MAE={mean_absolute_error(y_pred_rf,y_reg_test):.4f}')
print(f'随机森林:MSE={mean_squared_error(y_pred_rf,y_reg_test):.4f}')
print(f'随机森林:R²={r2_score(y_reg_test,y_pred_rf):.4f}')

# 4. 可视化
fig=plt.figure(figsize=(14,5))
ax1=plt.subplot(121) # 第一幅图: 预测值VS真实值散点图
ax2=plt.subplot(122) # 第二幅图: 特征重要性条形图

# 第一幅图: 预测值VS真实值散点图
sns.scatterplot(x=y_reg_test,y=y_pred_rf,ax=ax1,palette='husl',alpha=0.3)
ax1.plot([0,5],[0,5],ls='--',color='r',lw=2,label='理想预测线') # 对角参考线
ax1.set_xlabel('真实房价')
ax1.set_ylabel('预测房价')
ax1.set_title('随机森林:预测值VS真实值')
ax1.legend()

# 第二幅图: 特征重要性条形图
importance=pd.Series(best_rf.feature_importances_,index=X_reg.columns).sort_values()
sns.barplot(x=importance.values,y=importance.index,hue=importance.index,ax=ax2,p

```

Fitting 5 folds for each of 20 candidates, totalling 100 fits

RF最优参数: {'n_estimators': 300, 'min_samples_split': 5, 'min_samples_leaf': 1, 'max_features': 'log2', 'max_depth': None}

随机森林:MAE=0.3205

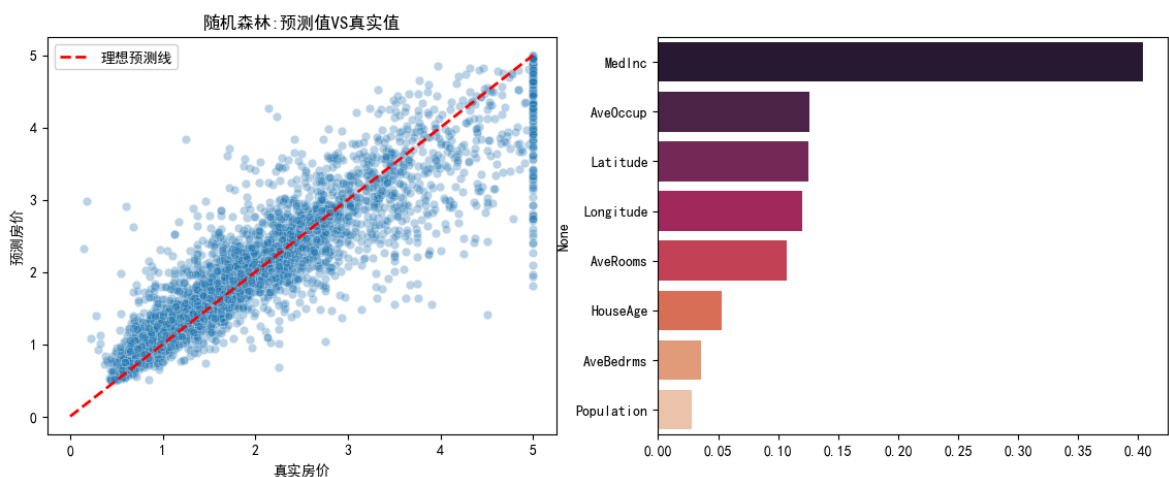
随机森林:MSE=0.2406

随机森林:R²=0.8164

C:\Users\11146\AppData\Local\Temp\ipykernel_36404\3604916734.py:42: UserWarning: Ignoring `palette` because no `hue` variable has been assigned.

```
sns.scatterplot(x=y_reg_test,y=y_pred_rf,ax=ax1,palette='husl',alpha=0.3)
```

出局[21]... <Axes: ylabel='None'>



3.XGBoost

在[22]中... `from xgboost import XGBRegressor`

```

# 1. 训练基础模型
xgb_reg=XGBRegressor(
    n_estimators=100,
    learning_rate=0.1,
    max_depth=6, # 单棵树深度

```

```

    subsample=0.8, # 每棵树使用的样本比例
    colsample_bytree=0.8, # 每棵树使用的特征比例
    random_state=RANDOM_STATE,
    n_jobs=-1
)
xgb_reg.fit(X_reg_train,y_reg_train)

# 2. 调参方法
# 关键超参数:@reg_alpha->L1正则项 @reg_lambda->L2正则项
xgb_params={
    'n_estimators':[100,300,500],
    'learning_rate':[0.01,0.05,0.1,0.2],
    'max_depth':[3,5,7],
    'subsample':[0.7,0.8,1.0],
    'colsample_bytree':[0.7,0.8,1.0],
    'reg_alpha':[0,0.1,1],
    'reg_lambda':[1,5,10]
}
xgb_search=RandomizedSearchCV(
    XGBRegressor(random_state=RANDOM_STATE,n_jobs=-1),
    xgb_params,
    cv=5,
    n_iter=20,
    scoring='neg_mean_squared_error',
    random_state=RANDOM_STATE,
    n_jobs=-1,
    verbose=1
)
xgb_search.fit(X_reg_train,y_reg_train)
print(f'XGB最优参数:{xgb_search.best_params_}')
best_xgb=xgb_search.best_estimator_
y_pred_xgb=best_xgb.predict(X_reg_test)

# 3. 评价指标
print(f'XGBoost:MAE={y_pred_xgb,y_reg_test}')
print(f'XGBoost:MSE={mean_squared_error(y_pred_xgb,y_reg_test):.4f}')
print(f'XGBoost:R²={r2_score(y_reg_test,y_pred_xgb):.4f}')

# 4. 可视化
fig=plt.figure(figsize=(14,5))
ax1=plt.subplot(121) # 第一幅图: 预测值VS真实值散点图
ax2=plt.subplot(122) # 第二幅图: 学习曲线: 训练过程中RMSE的变化(观察是否过拟合/欠拟合)

# 第一幅图: 预测值VS真实值散点图
sns.scatterplot(x=y_reg_test,y=y_pred_xgb,ax=ax1,alpha=0.3)
ax1.plot([0,5],[0,5],ls='--',color='r',lw=2,label='理想预测线') # 对角参考线
ax1.set_xlabel('真实房价')
ax1.set_ylabel('预测房价')
ax1.set_title('XGBoost: 预测值VS真实值')
ax1.legend()

# 第二幅图: 学习曲线图
eval_set=[(X_reg_train, y_reg_train), (X_reg_test, y_reg_test)]
best_xgb.fit(X_reg_train,y_reg_train,eval_set=eval_set,verbose=False) # verbose=
ax2.plot(best_xgb.evals_result()['validation_0']['rmse'],label='训练集RMSE')
ax2.plot(best_xgb.evals_result()['validation_1']['rmse'],label='测试集RMSE')
ax2.set_xlabel('迭代次数')
ax2.set_ylabel('RMSE')
ax2.set_title('XGBoost: 学习曲线')

```

```
ax2.legend()
plt.tight_layout()
```

Fitting 5 folds for each of 20 candidates, totalling 100 fits

XGB最优参数: {'subsample': 0.8, 'reg_lambda': 10, 'reg_alpha': 0, 'n_estimators': 500, 'max_depth': 7, 'learning_rate': 0.05, 'colsample_bytree': 0.7}

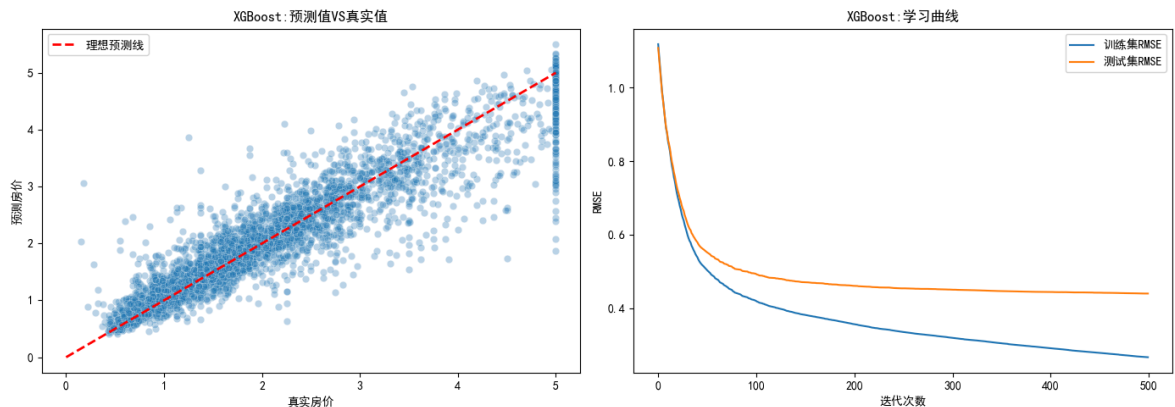
XGBoost:MAE=(array([0.523947 , 0.89055526, 5.163697 , ..., 4.979992 , 0.675313 35,

1.7720238], shape=(4128,), dtype=float32), array([0.477 , 0.458 , 5.000 01, ..., 5.00001, 0.723 , 1.515],

shape=(4128,)))

XGBoost:MSE=0.1931

XGBoost:R²=0.8526



4.轻型GBM

在[23]中...

```
from lightgbm import LGBMRegressor

# 1. 训练基础模型
lgbm_reg=LGBMRegressor(
    n_estimators=100,
    learning_rate=0.1,
    num_leaves=31, # LightGBM特有:叶子树(代替深度控制复杂度),即每棵树最多能有的
    subsample=0.8,
    colsample_bytree=0.8,
    random_state=RANDOM_STATE,
    n_jobs=-1,
    verbose=-1 # verbose=-1表示关闭日志
)
lgbm_reg.fit(X_reg_train,y_reg_train)
y_pred_lgbm=lgbm_reg.predict(X_reg_test)

# 2. 调参方法
# 核心参数:@num_Leaves @min_child_samples:叶子最小样本数(逐叶子分裂)(如果传入权重
lgbm_params={
    'estimators':[100,300,500],
    'learning_rate':[0.01,0.05,0.1,0.2],
    'num_leaves':[15,31,63,127],
    'min_child_samples':[5,10,20,50],
    'subsample':[0.7,0.8,1],
    'colsample_bytree':[0.7,0.8,1],
    'reg_alpha':[0,0.1,1],
    'reg_lambda':[0,1,10]
}
lgbm_search=RandomizedSearchCV(
    LGBMRegressor(random_state=RANDOM_STATE,n_jobs=-1,verbose=-1),
    lgbm_params,
    n_iter=20,
```

```

cv=5,
scoring='neg_mean_squared_error',
random_state=RANDOM_STATE,
n_jobs=-1,
verbose=1
)
lgbm_search.fit(X_reg_train,y_reg_train)
print(f'LGBM最优参数:{lgbm_search.best_params_}')
best_lgbm=lgbm_search.best_estimator_
y_pred_lgbm=best_lgbm.predict(X_reg_test)

# 3. 评价指标
print(f'LightGBM:MAE={y_pred_lgbm,y_reg_test}')
print(f'LightGBM:MSE={mean_squared_error(y_pred_lgbm,y_reg_test):.4f}')
print(f'LightGBM:R²={r2_score(y_reg_test,y_pred_lgbm):.4f}')

# 4. 可视化
fig=plt.figure(figsize=(14,5))
ax1=plt.subplot(121) # 第一幅图: 预测值VS真实值散点图
ax2=plt.subplot(122) # 第二幅图: 特征重要性排序图

# 第一幅图
sns.scatterplot(x=y_reg_test,y=y_pred_lgbm,ax=ax1,alpha=0.3)
ax1.plot([0,5],[0,5],ls='--',color='r',lw=2,label='理想预测线') # 对角参考线
ax1.set_xlabel('真实房价')
ax1.set_ylabel('预测房价')
ax1.set_title('LightGBM: 预测值VS真实值')
ax1.legend()

# 第二幅图
importance=pd.Series(best_lgbm.feature_importances_,index=X_reg.columns)
importance = importance / importance.sum()
importance = importance.sort_values(ascending=False)
sns.barplot(x=importance.values,y=importance.index,hue=importance.index,ax=ax2,p

```

Fitting 5 folds for each of 20 candidates, totalling 100 fits

LGBM最优参数: {'subsample': 0.8, 'reg_lambda': 10, 'reg_alpha': 1, 'num_leaves': 127, 'min_child_samples': 5, 'learning_rate': 0.1, 'estimators': 500, 'colsample_bytree': 0.7}

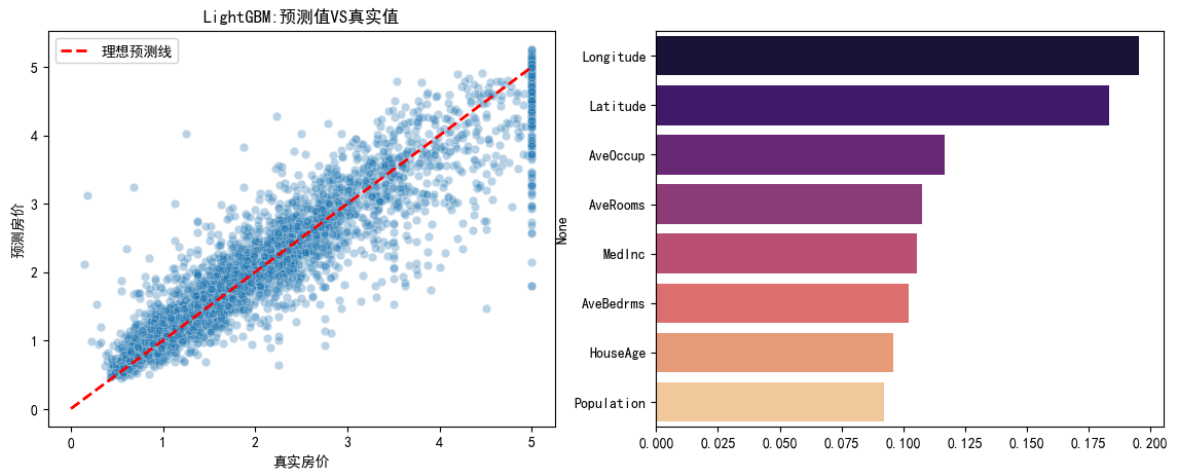
LightGBM:MAE=(array([0.56245953, 0.92455664, 4.88423656, ..., 4.99756581, 0.75384114,

1.74605805], shape=(4128,)), array([0.477 , 0.458 , 5.00001, ..., 5.00001, 0.723 , 1.515], shape=(4128,)))

LightGBM:MSE=0.1945

LightGBM:R²=0.8516

出局[23]... <Axes: ylabel='None'>



5. AdaBoost (自适应提升)

在[25]中...

```
from sklearn.ensemble import AdaBoostRegressor

# 1. 训练基础模型
ada_reg=AdaBoostRegressor(
    n_estimators=100,
    learning_rate=0.5,
    random_state=RANDOM_STATE
)
ada_reg.fit(X_reg_train,y_reg_train)

# 2. 调参方法
# AdaBoost调参重点:n_estimators与learning_rate(二者需联动,学习率小则需要更多轮数)
from sklearn.tree import DecisionTreeRegressor
ada_params = {
    'n_estimators': [50, 100, 200, 300],
    'learning_rate': [0.01, 0.1, 0.5, 1.0],
    'estimator': [
        DecisionTreeRegressor(max_depth=1), # 树桩(默认)
        DecisionTreeRegressor(max_depth=3), # 稍深的树
        DecisionTreeRegressor(max_depth=5)
    ]
}
ada_grid = GridSearchCV(
    AdaBoostRegressor(random_state=RANDOM_STATE),
    ada_params, cv=5, scoring='neg_mean_squared_error', n_jobs=-1
)
ada_grid.fit(X_reg_train, y_reg_train)
print("AdaBoost 最优参数:", ada_grid.best_params_)
best_ada = ada_grid.best_estimator_
y_pred_ada = best_ada.predict(X_reg_test)

# 3. 评价指标
print(f"[AdaBoost] MAE={mean_absolute_error(y_reg_test, y_pred_ada):.4f} | "
      f"RMSE={np.sqrt(mean_squared_error(y_reg_test, y_pred_ada)):.4f} | "
      f"R²={r2_score(y_pred_ada,y_reg_test):.4f}")

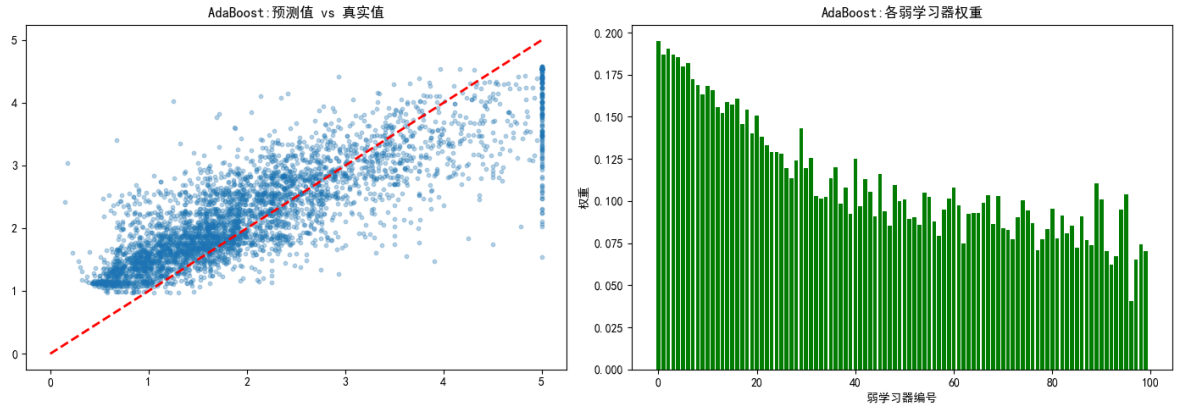
# 4. 可视化
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].scatter(y_reg_test, y_pred_ada, alpha=0.3, s=10)
axes[0].plot([0, 5], [0, 5], 'r--', lw=2)
axes[0].set_title('AdaBoost: 预测值 vs 真实值')
```



```
# 各弱学习器的权重分布
axes[1].bar(range(len(best_ada.estimator_weights_)), best_ada.estimator_weights_)
axes[1].set_xlabel('弱学习器编号'); axes[1].set_ylabel('权重')
axes[1].set_title('AdaBoost:各弱学习器权重')
plt.tight_layout(); plt.show()
```

AdaBoost 最优参数: {'estimator': DecisionTreeRegressor(max_depth=5), 'learning_rate': 0.1, 'n_estimators': 100}

[AdaBoost] MAE=0.5105 | RMSE=0.6553 | $R^2=0.3571$



6.回归任务:模型横向对比

在[26]中...

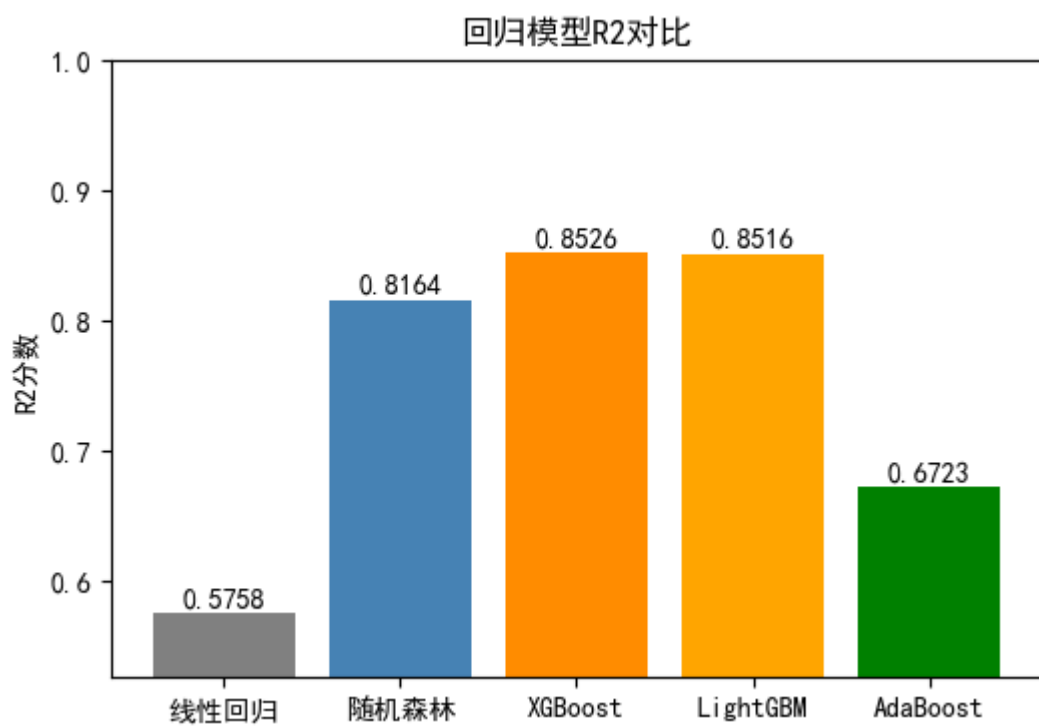
```
# 回归模型汇总对比
reg_results = pd.DataFrame({
    '模型': ['线性回归', '随机森林', 'XGBoost', 'LightGBM', 'AdaBoost'],
    'MAE': [mean_absolute_error(y_reg_test, y_pred_lr),
            mean_absolute_error(y_reg_test, y_pred_rf),
            mean_absolute_error(y_reg_test, y_pred_xgb),
            mean_absolute_error(y_reg_test, y_pred_lgbm),
            mean_absolute_error(y_reg_test, y_pred_ada)],
    'RMSE': [np.sqrt(mean_squared_error(y_reg_test, p)) for p in
             [y_pred_lr, y_pred_rf, y_pred_xgb, y_pred_lgbm, y_pred_ada]],
    'R²': [r2_score(y_reg_test, p) for p in
           [y_pred_lr, y_pred_rf, y_pred_xgb, y_pred_lgbm, y_pred_ada]]
})
reg_results
```

出局[26]...

	模型	MAE	RMSE	R^2
0	线性回归	0.533199	0.745579	0.575791
1	随机森林	0.320469	0.490475	0.816419
2	XGBoost	0.286792	0.439481	0.852608
3	LightGBM	0.286619	0.441012	0.851579
4	阿达增强	0.510545	0.655345	0.672257

在[27]中...

```
# 柱状图对比  $R^2$ 
plt.figure(figsize=(6, 4))
plt.bar(reg_results['模型'], reg_results['R²'],
        color=['gray', 'steelblue', 'darkorange', 'orange', 'green'])
plt.ylabel('R2分数'); plt.title('回归模型R2对比')
plt.ylim(min(reg_results['R²']) - 0.05, 1.0)
for i, v in enumerate(reg_results['R²']):
    plt.text(i, v + 0.005, f'{v:.4f}', ha='center')
plt.show()
```



在[]中：